

County-Level Corn and Cotton Yield Prediction in Texas

Polished project report generated from the current workspace run

Syed Ali | Wajeeh Moonwar | Gustavo Hernandez

Texas State University

Project Focus	Corn and Cotton yield forecasting at the Texas county level
Core Model	XGBoost regressors with rolling-origin validation
Weather Source	NASA POWER REST API, mapped from county coordinates
Crop and Soil Source	USDA NASS yield records and USDA soil data

Abstract

This report summarizes a walk-forward XGBoost pipeline for county-level Corn and Cotton yield prediction in Texas. The workflow combines USDA NASS yield records, USDA soil variables, and NASA POWER meteorological data. A geospatial preprocessing pipeline was required to map each county coordinate to the NASA POWER REST API so that weather observations could be retrieved consistently for every county-year pair.

Purpose and Process

- Establish a reproducible agricultural modeling workflow for two major Texas crops.
- Join crop, soil, and weather information into biologically complete county-year records.
- Engineer seasonally aligned features such as drought stress, moisture balance, and vapor pressure deficit proxies.
- Evaluate predictive skill with rolling-origin validation rather than a single random train-test split.

Data Sources

- NASA POWER weather data collected through REST API calls after geospatial county-coordinate mapping.
- USDA NASS county-level crop yield data for Corn and Cotton.
- USDA soil data for static county soil attributes such as clay percentage and pH.

Modeling Approach

- XGBoost regressors were trained separately for Corn and Cotton.
- Rolling-origin validation started at the first out-of-time year to mimic forecasting conditions.

- Engineered features captured crop-specific seasonal stress rather than relying only on raw monthly averages.

Key Results

Crop	Cumulative R2	RMSE	MAE	Notes
Corn	0.6710	24.69 bu/acre	19.54 bu/acre	Strongest and most stable performance across multiple test years
Cotton	0.4948	248.62 lbs/acre	209.17 lbs/acre	Only one out-of-time test year in the current workflow

Corn outperformed Cotton in both average accuracy and stability. The Corn model reached R2 values above 0.80 in 2021 and remained above 0.65 in several other years. Cotton remained predictive but was more difficult to fit, which is consistent with the crop's indeterminate growth habit and the smaller number of fully merged county-year records.

Comparison to the Draft Report

The executed pipeline produced results that are close to the values reported in the draft analysis, but not identical. Corn remained in the same general performance band, while Cotton showed a larger drop in R2 on the local run. The discrepancy is consistent with data-split sensitivity and the limited number of Cotton validation years.

Conclusion

The project demonstrates that an interpretable machine learning workflow can capture a substantial share of county-level yield variation when the data pipeline respects crop biology, geography, and seasonality. The strongest improvement opportunity is Cotton, where broader historical coverage and more deterministic preprocessing would likely improve validation reliability.

Generated from the current workspace execution of code.py and aligned with the updated markdown summary.